

PredictFun: Bootstrapping Prediction Markets With Virtual LMSR And Band-Pass Graduation Clearing

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June 2026

WORKING DRAFT, rev. 0.4

Abstract

A new prediction market faces a bootstrapping dilemma: it needs a credible price before it has liquidity, but in a thin market every trade moves the price violently. Subsidized market makers solve this for a handful of flagship markets and fail for the long tail. This paper describes a launch mechanism that needs no subsidy at all. Before graduation, each market runs a virtual Logarithmic Market Scoring Rule (LMSR): the liquidity parameter b shapes how smoothly the price moves but is backed by no bankroll, and every pre-graduation bet is a priced intent — a receipt recording the exact segment of the price curve it traversed — rather than a final outcome token.

Graduation converts intents into a standard fixed-payout market in the style of Polymarket's Conditional Token Framework (CTF). A band-pass clearing rule passes exactly the price bands traversed by both YES and NO demand in opposite directions. Matched segments mint fully collateralized YES/NO complete sets, every unmatched segment is refunded at the exact path cost the bettor paid for it, and the locked collateral provably equals the maximum payout of the claims it backs.

The tradeoffs are explicit rather than hidden. Pre-graduation capital is locked until the market clears or expires, fills are partial and never guaranteed, and a bettor's worst case is a full refund. In exchange, a market can open at any prior probability, discover prices smoothly from its first trade, and hand the discovered demand into ordinary prediction-market infrastructure the moment enough real opposing interest exists.

1. The Bootstrapping Problem

A new prediction market needs a price before it has liquidity, but without liquidity it cannot discover a price worth trusting. When the book is thin, a small order moves the displayed odds dramatically, so the price reflects order size more than belief. Traders who could correct the price stay away precisely because the price is noisy, and the market never escapes the loop.

The direct solution is subsidy: seed each market with enough market-maker capital that early trades do not move the price too much. This works for a small number of important markets and scales poorly past them. A protocol that launches many markets must either hold a large treasury, accept shallow prices, or burn scarce capital on markets that never attract organic demand. It may also simply not have the money.

A central limit order book solves a different part of the problem. It gives precise matching once makers and takers arrive, but it does not give immediate price discovery. An empty or thin book is not a smooth probability curve; it is a waiting room for limit orders.

LMSR comes closest to the desired launch primitive. It quotes from its first block, moves smoothly under sparse flow, and expresses probabilities directly rather than through reserve ratios. The catch is that conventional LMSR is not free: its liquidity parameter b is tied to a funded market-maker loss budget, so every new market would again need a subsidy.

This paper keeps LMSR's price-discovery surface and removes its funding requirement by changing what a pre-graduation trade is. The parameter b becomes virtual smoothness rather than funded inventory, pre-graduation bets become priced intents rather than final token purchases, and graduation becomes the explicit step that decides which intents can become fixed-payout claims.

Stage 1 — Bootstrap: a virtual LMSR prices intents; no subsidy, no final fills.

Stage 2 — Graduation: band-pass clearing converts path-compatible intents into complete sets.

Stage 3 — Standard market: CTF-style YES/NO tokens trade on an ordinary venue.

The third stage is the point. The virtual curve is scaffolding: it exists to discover a price and gather committed demand for a market that does not yet deserve a funded order book, and it is designed to retire itself the moment the market does.

2. Background: The Destination And The Candidates

It helps to fix the destination before judging launch mechanisms, because the destination is what makes launch hard.

Fixed-Payout Claims And The CTF Destination

A binary prediction market outcome token has a fixed terminal payoff:

1 YES $\mapsto$ 1 unit of collateral	if YES resolves true
1 NO $\mapsto$ 1 unit of collateral	if NO resolves true

For a collateral-complete binary market:

$$1 \text{ YES} + 1 \text{ NO} = 1 \text{ unit of collateral}$$

This constraint is what separates prediction-market launch from token launch. A token bonding curve can sell an asset with no fixed redemption value. A prediction market cannot safely sell 100 final YES claims at a price of 0.05, collect 5 units of collateral, and later owe 100 units if YES wins — unless something else funds the missing 95.

Fully collateralized outcome-token frameworks are the right destination once liquidity exists. Polymarket's Conditional Token Framework (CTF), built on Gnosis Conditional

Tokens, is the reference example: YES and NO are ordinary tokens backed by complete sets of collateral, tradable on order books and standard venues. The bootstrapping problem is not that this destination is wrong. It is that a brand-new market must discover a price and attract opposing demand before there are any complete sets worth trading.

Pari-Mutuel Pools

Pari-mutuel markets stay solvent by changing the payoff: winners split the pool after betting closes. No subsidy is needed, but the bettor never holds a fixed-payout claim at a known price — final odds depend on the final pool. As a launch mechanism it gives weaker price certainty than the tokenized product users expect, and it never converts into one.

Reserve-Based Market Makers

Constant-product and fixed-product market makers quote from reserves. In a binary fixed-product market, the approximate YES price is:

$$P_{\text{yes}} = \frac{B_{\text{no}}}{B_{\text{yes}} + B_{\text{no}}}$$

A 5% YES quote therefore requires:

$$B_{\text{yes}} \approx 19 B_{\text{no}}$$

If those reserves are real, the market opens with heavily imbalanced inventory and liquidity providers carry outcome risk before demand is proven. If the reserves are virtual, the displayed depth is not collateral that can pay winners. Reserve curves are useful once a market has committed liquidity; they do not solve no-subsidy launch.

LMSR

For a binary market, LMSR prices trades from the cost function:

$$C(q_{\text{yes}}, q_{\text{no}}) = b \ln \left(e^{q_{\text{yes}}/b} + e^{q_{\text{no}}/b} \right)$$

The marginal YES price is:

$$P_{\text{yes}} = \frac{e^{q_{\text{yes}}/b}}{e^{q_{\text{yes}}/b} + e^{q_{\text{no}}/b}} \quad P_{\text{no}} = 1 - P_{\text{yes}}$$

and the cost of buying s YES shares is:

$$\text{cost}_{\text{yes}} = C(q_{\text{yes}} + s, q_{\text{no}}) - C(q_{\text{yes}}, q_{\text{no}})$$

LMSR is attractive for sparse launch markets because its state is a probability score, not reserve inventory. It can open at a meaningful prior, quote continuously, and move smoothly as demand arrives. The cost is the classical funding requirement: a binary LMSR market maker that sells final tokens can lose up to

$$\max \text{loss} = b \ln 2$$

and someone must bankroll that bound. If nobody does, LMSR cannot be treated as a seller of final fixed-payout claims.

Comparison

Mechanism	Strength at launch	Weakness at launch
Subsidized LMSR	Instant smooth quotes and guaranteed fills	Needs a funded loss budget per market
Pari-mutuel pool	Solvent with zero subsidy	No fixed payout at entry odds; never becomes CTF-style
Reserve AMM (CPMM)	Familiar passive liquidity	Real reserves take outcome risk; virtual reserves cannot pay winners
Central limit order book	Precise matching at scale	An empty book discovers nothing
Virtual LMSR + band-pass clearing	Immediate discovery, no subsidy, solvent graduation	Fills deferred, partial, and not guaranteed; capital locked until clearing

The last row is this paper. The rest of the paper makes its two halves precise: how a virtual LMSR prices intents, and how band-pass clearing decides which intents become real.

3. Virtual LMSR: Price Discovery Without A Bankroll

What The State Means

Before graduation, the market runs full LMSR math, but the meaning of the state changes:

q_{yes} and q_{no} are demand-pricing state, not sold inventory.
 b is virtual smoothness, not a funded loss budget.
every bet is a priced intent; its collateral stays escrowed.
no final YES/NO token exists before graduation clearing.

Because no final token is sold, no loss bound needs funding. The curve can be arbitrarily smooth — b is a free design parameter per market — without anyone underwriting $b \ln 2$.

Starting Away From Even Odds

Markets often should not open at 50/50: a longshot event may deserve a 5% prior, a near-consensus event an 80% prior. LMSR handles this with a pure state offset. Define the path coordinate and its price:

$$\begin{aligned} r &= q_{\text{yes}} - q_{\text{no}} \\ P_{\text{yes}}(r) &= \frac{1}{1 + e^{-r/b}} \\ P_{\text{no}}(r) &= 1 - P_{\text{yes}}(r) \end{aligned}$$

To open at probability P_0 , initialize the state at:

$$r_0 = b \ln \left(\frac{P_0}{1 - P_0} \right)$$

For a 5% opening probability:

$$r_0 = b \ln \left(\frac{0.05}{0.95} \right) \approx -2.9444 b$$

The offset is a coordinate, not collateral. The market can quote 5% without pretending anyone holds a deep reserve of cheap YES claims — something a reserve-based AMM cannot do without either real imbalanced inventory or unbacked virtual depth.

What The Quote Promises

The displayed price is real information but a limited promise. It is an implied probability, and it is the exact pricing rule applied to the next intent. It is not a fill: quoting 35% does not mean anyone can settle exposure at 35%, because no opposing collateral has been proven to exist there.

This is the central technical problem the rest of the paper solves. A virtual LMSR can quote any price without evidence of opposite-side demand at that price. If the protocol treated virtual-LMSR buys as final tokens, it would silently reintroduce the missing subsidy as bad debt. The mechanism instead records, for every bet, exactly which part of the curve it bought — and defers the question of what is real to graduation:

Did YES and NO demand traverse the same price bands in opposite directions?

Only the segments where the answer is yes become fixed-payout exposure.

4. Receipts: Priced Intents Over A Price Range

The Receipt Record

Each pre-graduation buy creates a receipt — the durable record of one priced intent:

owner_ℓ : receipt holder
side_ℓ ∈ {YES, NO}
s_ℓ : provisional shares
c_ℓ : escrowed cost basis
(q_{yes}⁰, q_{no}⁰)_ℓ : LMSR state before the trade
(q_{yes}¹, q_{no}¹)_ℓ : LMSR state after the trade
τ_ℓ : sequence index

A YES buy of s shares raises q_{yes} by s, so it raises r by s; a NO buy lowers r by s. Every receipt therefore traverses an interval on the path coordinate whose width equals its share count:

$$\begin{aligned} r_{\text{low},\ell} &= \min(r_\ell^0, r_\ell^1) \\ r_{\text{high},\ell} &= \max(r_\ell^0, r_\ell^1) \\ r_{\text{high},\ell} - r_{\text{low},\ell} &= s_\ell \end{aligned}$$

Path Cost

A receipt's cost basis is the integral of the relevant marginal price over its interval. Both integrals have closed forms:

$$\begin{aligned}\text{YES_cost}[u, v] &= \int_u^v P_{\text{yes}}(r) dr = b \ln \left(\frac{1 + e^{v/b}}{1 + e^{u/b}} \right) \\ \text{NO_cost}[u, v] &= \int_u^v P_{\text{no}}(r) dr = b \ln \left(\frac{1 + e^{-u/b}}{1 + e^{-v/b}} \right)\end{aligned}$$

In probability terms the forms are even simpler. If a segment runs from displayed probability P_a to P_b , then:

$$\begin{aligned}\text{YES_cost} &= b \ln \left(\frac{1 - P_a}{1 - P_b} \right) \\ \text{NO_cost} &= b \ln \left(\frac{P_b}{P_a} \right)\end{aligned}$$

Every number in this paper's worked examples can be reproduced from these two formulas. A receipt's average price,

$$p_\ell = \frac{c_\ell}{s_\ell}$$

is useful for display, but it deliberately plays no role in clearing: the cost basis is distributed unevenly along the interval, because the marginal price changes as the trade moves the curve. The early slices of a YES buy are cheaper than the late slices, and clearing must respect that.

Buying A Range

It is worth saying plainly what a pre-graduation buyer is buying, because it is not a token at one price. A receipt is a priced intent over a range of odds. The trader who moves the YES price from 20% to 40% has bought a continuous strip of provisional shares — one infinitesimal slice at every price between 20% and 40% — and paid for each slice exactly the marginal price at which it moved the curve. The receipt behaves like a dense strip of tiny limit orders, one resting at every price in the range, except that none of them has filled yet.

Graduation later decides, slice by slice, which parts of the strip became real. Every slice ends in exactly one of two states: converted into a fixed-payout share at the price recorded for that slice, or refunded at the price recorded for that slice. Nothing is ever settled at an average that mixes slices together, and nothing is ever kept without the collateral to pay it.

Entering By Budget

A trader who specifies a budget c instead of a share count gets the share count by inverting the cost function. With:

$$A = e^{q_{\text{yes}}/b}, \quad B = e^{q_{\text{no}}/b}$$

a YES budget c buys:

$$s = b \ln \left(\frac{e^{c/b}(A + B) - B}{A} \right)$$

and a NO budget, symmetrically:

$$s = b \ln \left(\frac{e^{c/b}(A + B) - A}{B} \right)$$

What A Receipt Grants

A receipt gives its holder exactly three rights:

1. The right to participate in deterministic graduation clearing.
2. The right to receive fixed-payout outcome tokens for every matched path segment, at that segment's recorded cost.
3. The right to a refund of every unmatched segment's recorded cost.

A receipt does not give its holder:

- A guaranteed fill, in whole or in part.
- A final YES or NO position before clearing.
- The ability to withdraw, transfer, or sell the receipt before clearing or cancellation.
- Any resolution payout before graduation.

The withdrawal restriction is deliberate, and it is the price of making intents meaningful. The displayed curve is the cumulative path of every escrowed intent. If a trader could move the price and then exit freely, the bootstrap curve would become a costless manipulation surface: pump the displayed odds, attract the other side, withdraw. Escrowed, append-only intents make every displayed price a price someone actually paid and committed. Locked receipts also keep clearing deterministic — the algorithm replays a frozen book, so every implementation reaches the same result.

The cost of this rule is stated honestly: pre-graduation capital is committed from the moment of purchase until the market clears, is canceled, or reaches its close time and refunds. Section 8 quantifies exactly what the committed capital can turn into.

5. Why Aggregate Matching Fails

Receipts record paths because nothing coarser is safe to clear. Several tempting shortcuts look solvent in aggregate and fail at the level fixed payouts require.

An Unfunded Loss Bound

The classical LMSR bound $b \ln 2$ is a funding requirement, not a solvency proof. Selling final tokens from a virtual LMSR and pointing at the bound just hides the subsidy assumption. A no-subsidy protocol cannot pay winners with a bound.

Global Share Proration

Equal total YES and NO share counts do not imply the two sides met. A YES wave from 5% to 20% and a NO wave from 80% to 65% can have identical widths while sharing no price

band. Retaining both because the totals balance would mint claims from demand that never opposed each other.

Global Collateral Proration

Matching total YES dollars against total NO dollars fails the same way for a different reason: a few high-price shares can contribute the same collateral as many low-price shares while creating a very different payout liability. Fixed-payout solvency is a constraint on retained share count, not retained dollars.

Average-Price Partial Fills

A YES receipt that moved the curve from 50% to 60% paid cheap marginal prices near 50% and expensive ones near 60%. If only its 55% to 60% segment finds opposing demand, the retained exposure costs what that upper segment cost — not the receipt's 55.03% average. Settling partial fills at average price over-refunds one segment, under-collects another, and breaks the collateral accounting band by band.

Virtual Reserves

A pre-graduation CPMM with virtual reserves can imitate depth, but virtual liquidity cannot redeem winning claims. Some later rule must still decide which exposure is backed by real collateral. Band-pass clearing is that rule; virtual-reserve accounting only postpones the question.

What survives all five objections is the path itself. The clearing rule must compare where YES demand actually traveled against where NO demand actually traveled, at full resolution.

6. Band-Pass Graduation Clearing

Band-pass clearing treats the price path as the object to be matched, and passes exactly the bands crossed by both sides — by analogy with a band-pass filter, which passes a band of frequencies and rejects the rest.

The Sweep

Clearing runs over a frozen receipt book. Each receipt is already an interval on r , so the algorithm is a deterministic sweep:

1. Convert every active receipt into its interval $[r_{\text{low}}, r_{\text{high}}]$.
2. Collect all interval endpoints, sort them, and deduplicate.
3. Sweep each adjacent band $[r_k, r_{k+1}]$, skipping zero-width and dust-width bands.
4. Find the YES receipts whose intervals cover the whole band; count them as Y_k .
5. Find the NO receipts whose intervals cover the whole band; count them as N_k .
6. If either count is zero, the band fails: no exposure in it is retained.
7. Otherwise retain the scarce side fully and the crowded side pro-rata, using $m_k = \min(Y_k, N_k)$.
8. Credit each covering receipt with its retained band shares and retained band cost.
9. After the sweep, refund every receipt's unretained path cost.
10. Sum the matched market cap F and apply the graduation decision of Section 7.

Because the endpoint set includes every receipt boundary, each band is either fully inside or fully outside any given receipt — no partial coverage cases exist.

The sweep is indifferent to time. A receipt matches any opposite-side receipt that covered the same band, whether it traded earlier or later; sequence indices exist for deterministic ordering and rounding tie-breaks, not priority. A FIFO variant — earlier receipts matched first within a band — would be a coherent alternative allocation policy, but it is a policy choice layered on top of the same solvency math. This paper analyzes pro-rata matching because solvency depends only on path overlap, while priority is negotiable.

Band Arithmetic

For each band define:

$$\begin{aligned} w_k &= r_{k+1} - r_k \\ Y_k &= \#\{\text{YES receipts covering } [r_k, r_{k+1}]\} \\ N_k &= \#\{\text{NO receipts covering } [r_k, r_{k+1}]\} \\ m_k &= \min(Y_k, N_k) \end{aligned}$$

The retained fractions are:

$$\text{YES_fraction}_k = \frac{m_k}{Y_k} \quad \text{NO_fraction}_k = \frac{m_k}{N_k}$$

Each matched band adds, to every YES receipt covering it:

$$\begin{aligned} \Delta \text{retained_shares}_\ell &= \frac{m_k}{Y_k} w_k \\ \Delta \text{retained_cost}_\ell &= \frac{m_k}{Y_k} \text{YES_cost}[r_k, r_{k+1}] \end{aligned}$$

and symmetrically for NO receipts with N_k and NO_cost . The fixed-payout market cap contributed by the band is:

$$F_k = w_k m_k$$

Proration scales a receipt's band shares and band cost by the same fraction, so it changes how much of a band a receipt keeps — never the per-share price of what it keeps.

Local Collateral Completeness

Solvency comes from the complementarity of binary LMSR prices. For any band:

$$\begin{aligned} \text{YES_cost}[u, v] + \text{NO_cost}[u, v] &= \int_u^v (P_{\text{yes}}(r) + P_{\text{no}}(r)) dr \\ &= \int_u^v 1 dr \\ &= v - u \end{aligned}$$

The band's width is also the number of YES/NO complete sets it represents. So when one YES segment and one NO segment overlap a band, their combined recorded cost is exactly the collateral that fully backs the band's complete sets — one unit per set, locally, with nothing borrowed from any other band. The collateral locked by band k is:

$$L_k = m_k (\text{YES_cost}[r_k, r_{k+1}] + \text{NO_cost}[r_k, r_{k+1}]) = m_k w_k = F_k$$

Completeness is proven band by band; no global argument is needed, and no band can be solvent at another band's expense.

The Conservation Identity

Let:

E : total escrowed receipt collateral
 R : total refunds
 L : collateral locked behind retained claims
 F : retained fixed-payout market cap

Summing the band identity over all matched bands and refunding everything else:

$$L = \sum_k L_k = \sum_k F_k = F \quad \text{and} \quad E = R + L = R + F$$

Since binary fixed-payout settlement pays at most one unit per complete set:

$$\text{maximum winner payout} = F = L$$

The retained market is fully collateralized by construction — no subsidy, creator bond, insurance fund, or future fee revenue is ever needed to pay winners. Every individual receipt simultaneously satisfies:

$$\begin{aligned} \text{retained_cost}_\ell + \text{refund}_\ell &= c_\ell \\ 0 \leq \text{refund}_\ell &\leq c_\ell \\ \text{retained_cost}_\ell &\leq \text{retained_shares}_\ell \end{aligned}$$

The first line is cost-basis preservation: clearing splits a receipt's escrow between tokens and refund without creating or destroying a single unit. This mechanism charges no fees; if fees are ever added they must appear explicitly in the identity as $E = R + L + \text{fees}$, never implicitly.

7. Graduation: The Bridge To A Standard Market

Why Leave The Curve

Graduation is the point of the design, not a cosmetic milestone. The virtual LMSR exists to bootstrap: it gives a new market continuous odds before the market has earned conventional liquidity. Once enough path-compatible demand exists, the market should leave the virtual phase as fast as possible, because a CTF-style market is simply better once it has liquidity: claims are ordinary collateralized tokens, market makers can quote them, venues can list them, and traders no longer need to reason about provisional intents. Keeping a market on the virtual curve after it can support fixed-payout exposure would preserve bootstrapping complexity after the bootstrap has done its job.

Threshold And Non-Graduation

Graduation is decided by the one number clearing actually proves — the path-compatible filled market cap:

$$F = \sum_{\text{cleared bands } k} F_k \quad F \geq \text{graduation_threshold}$$

F is deliberately not provisional volume, headline open interest, total escrow, or the final displayed price. All of those can be large while path-compatible demand is near zero.

The lifecycle is short and total:

open for intents -> frozen for clearing -> graduated: tokens minted, residual refunds paid

open for intents -> close time reached below threshold -> not graduated: every receipt refunded in full

Once a market becomes eligible, anyone may freeze it; clearing then runs over the frozen book. If the close time arrives first, the no-subsidy rule is unconditional:

$$\text{retained_shares}_\ell = 0 \quad \text{refund}_\ell = c_\ell$$

for every receipt. Sub-threshold overlap may be mathematically solvent, but the threshold defines the minimum viable fixed-payout market: graduating tiny matched books would produce fragile markets whose apparent precision exceeds their real liquidity. The threshold can be paired with non-price gates — minimum unique participants, concentration limits — but those sit outside the clearing math; the clearing primitive remains F .

The Handoff

Graduation outputs exactly what a standard venue needs and nothing it must apologize for:

retained YES balances and retained NO balances
locked collateral equal to one unit per complete set
market metadata and resolution rules

Retained exposure can then trade on any venue that preserves the complete-set invariant — a Polymarket-style central limit order book, a CTF-compatible AMM, or a hybrid. The venue inherits fully backed tokens; it is never asked to rescue undercollateralized receipts. Post-graduation venue design and resolution are modular layers outside this paper's scope.

8. Fill Outcomes: Best Case And Worst Case

Everything a pre-graduation bettor can experience follows from the clearing rule. This section states it from the bettor's side, because "you bought an intent over a range" only becomes fair when the range of outcomes is explicit.

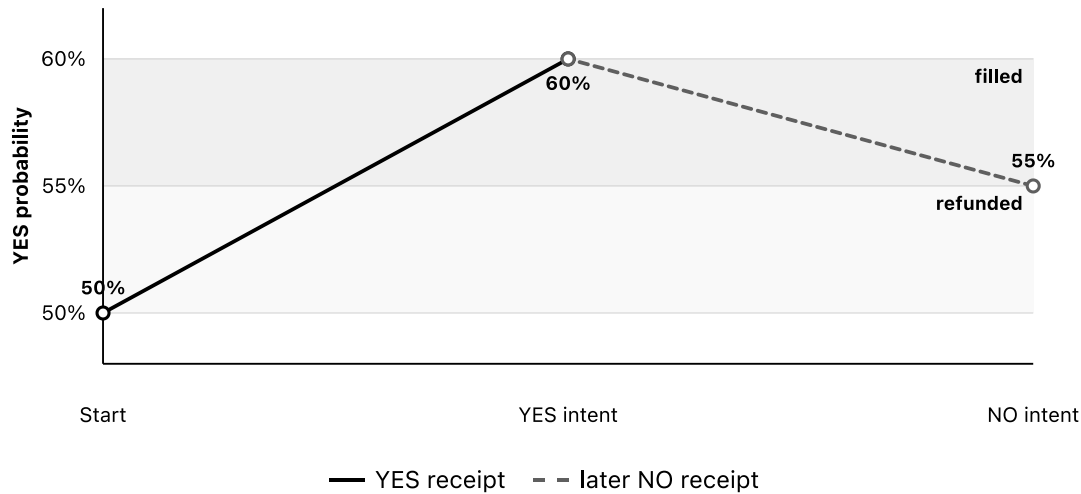
Four Outcomes

A receipt with shares s and escrowed cost c ends in exactly one of four ways:

Outcome	Condition	Retained shares	Refund
No graduation	market never reaches the threshold	0	c (full)
Graduated, no overlap	no band of the receipt's range cleared	0	c (full)
Graduated, partial fill	some bands cleared, or the receipt's side was crowded	between 0 and s	c minus retained cost
Graduated, full fill	every band cleared and the receipt's side was scarce throughout	s	0

The worst case is a full refund: the bettor recovers every escrowed unit and has paid only the time value of locking it. The best case is a full fill of s fixed-payout shares at the exact prices originally paid. Every intermediate case is a per-band mixture of those two, never a blend at made-up prices.

One Receipt, Two Fates



A YES receipt spans 50% to 60%; opposing demand retraces only the top half. Each half settles by its own recorded path cost: the 55% to 60% slice becomes fixed-payout shares, the 50% to 55% slice becomes a refund.

Quantity Is Uncertain, Price Is Not

The retained share count, for a YES receipt covering cleared-band set K , is:

$$\text{retained_shares}_\ell = \sum_{k \in K} \frac{m_k}{Y_k} w_k \quad 0 \leq \text{retained_shares}_\ell \leq s_\ell$$

Each term moves with the behavior of other traders: which bands the other side retraced, and how crowded the receipt's own side was in each. What never moves is price. The per-share price of every band a receipt keeps was fixed by the receipt's own path integral at trade time. Later traders decide which slices of the strip fill and in what fraction; they cannot make any filled slice more expensive.

Price Bounds For Partial Fills

A partial fill changes a receipt's effective average price, because the matched bands are not the average bands. The movement is strictly bounded. For a YES receipt over $[u, v]$ with any nonzero fill:

$$P_{\text{yes}}(u) \leq \frac{\text{retained_cost}_\ell}{\text{retained_shares}_\ell} \leq P_{\text{yes}}(v)$$

The effective price of the retained shares always lies between the marginal prices at the two ends of the range the bettor chose to sweep — never outside it. (For a NO receipt, the same bound holds with P_{no} and the endpoints exchanged.) The worst-case fill price is the top of the bettor's own range; the best case is the bottom. Matching is time-symmetric, so both cases are reachable: a receipt can match opposing flow that crossed its lower bands before it traded, or opposing flow that retraces its upper bands afterward.

Payoff Bounds

Let ret be retained shares and rc the retained cost, so the refund is $c - \text{rc}$. Because the marginal price is strictly below one at every finite r :

$$\text{retained_cost}_\ell < \text{retained_shares}_\ell \quad \text{whenever } \text{retained_shares}_\ell > 0$$

If the receipt's side wins, every retained share pays one unit:

$$\begin{aligned} \text{total received}_\ell &= \underbrace{(c_\ell - \text{retained_cost}_\ell)}_{\text{refund}} + \underbrace{\text{retained_shares}_\ell}_{\text{payout}} \\ &= c_\ell + (\text{retained_shares}_\ell - \text{retained_cost}_\ell) \\ &> c_\ell \end{aligned}$$

A winning bettor profits on any nonzero fill, and the profit per retained share is one minus that share's locked-in price. If the receipt's side loses:

$$\text{total received}_\ell = c_\ell - \text{retained_cost}_\ell \quad \text{maximum loss}_\ell = \text{retained_cost}_\ell \leq c_\ell$$

A bettor can lose only the cost of shares that actually filled — never escrow that didn't fill, and never more than was escrowed. There is no scenario with negative refund, no socialized loss, and no claim paid from anyone else's principal.

9. Worked Examples

Both examples use a single continuous price history, because that is what an LMSR produces: disjoint moves like "YES bought 5% to 20%" and "NO bought 80% to 65%" cannot coexist without the path traversing everything between them. Prices are quoted as probabilities; costs are in collateral units.

Example A: Three Traders, One Graduation

Let $b = 100$ and let the market open at 20%:

$$r_0 = 100 \ln \left(\frac{0.20}{0.80} \right) \approx -138.63$$

Three traders act in sequence:

Receipt	Side	Price path	Shares s	Cost c	Average price
Alice	YES	20% $\rightarrow$ 40%	98.08	28.77	0.2933
Noah	NO	40% $\rightarrow$ 30%	44.18	28.77	0.6511
Bea	YES	30% $\rightarrow$ 35%	22.83	7.41	0.3247

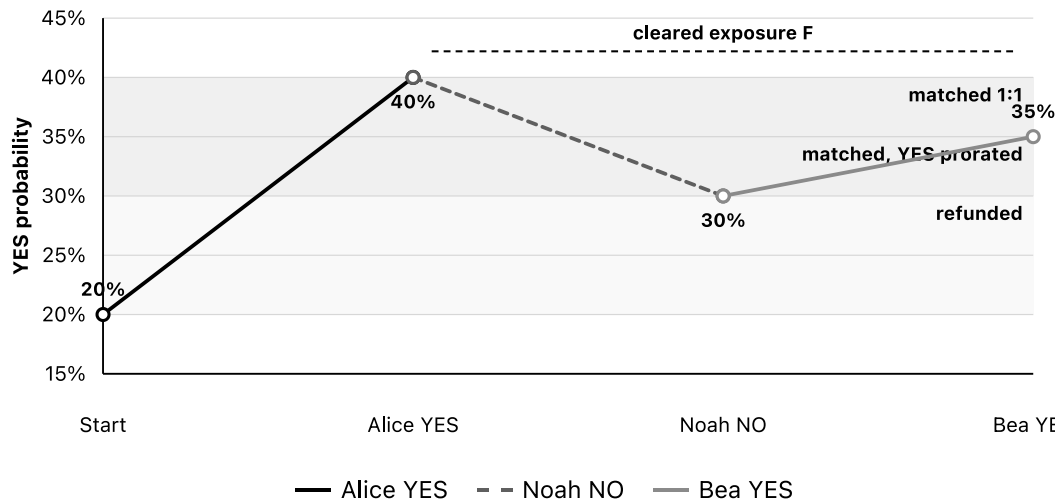
Every entry follows from Section 4's closed forms. Alice's cost, for instance:

$$c_{\text{Alice}} = b \ln \left(\frac{1 - P_a}{1 - P_b} \right) = 100 \ln \left(\frac{0.80}{0.60} \right) = 100 \ln \frac{4}{3} \approx 28.77$$

and Noah's, on the NO side over his own path:

$$c_{\text{Noah}} = b \ln \left(\frac{P_b}{P_a} \right) = 100 \ln \left(\frac{0.40}{0.30} \right) \approx 28.77$$

Example A Price Path



Noah's NO path retraces the top half of Alice's YES path, so 30% to 40% clears and the market ends at 35%. Alice and Bea both cover 30% to 35%, so the YES side of that band is prorated. Alice's 20% to 30% segment is never retraced and refunds in full.

The receipt endpoints split the path into three bands. Costs per band come from the same closed forms (for example, the YES cost of the 30%–35% band is $100 \ln(0.70/0.65) \approx 7.41$):

Band	Width w_k	YES covering	NO covering	m_k	YES fraction	NO fraction	F_k
20% – 30%	53.90	Alice	—	0	—	—	0
30% – 35%	22.83	Alice, Bea	Noah	1	1/2	1	22.83
35% – 40%	21.36	Alice	Noah	1	1	1	21.36

The 20%–30% band is one-sided and fails. The 30%–35% band has two YES receipts against one NO receipt, so Noah is retained fully while Alice and Bea each keep half of the band. The 35%–40% band matches one-to-one. Note the time symmetry: Noah traded once, yet his path matches Alice (who traded before him) and Bea (who traded after him) — coverage, not order, is what clears.

Accumulating per receipt:

$$\text{retained_shares}_{\text{Alice}} = \frac{1}{2}(22.8259) + 21.3574 = 32.7704$$

$$\text{retained_cost}_{\text{Alice}} = \frac{1}{2}(7.4108) + 8.0043 = 11.7097$$

$$\text{refund}_{\text{Alice}} = 28.7682 - 11.7097 = 17.0585$$

Trader	Retained shares	Retained cost	Refund	Effective price
Alice	32.77 YES	11.71	17.06	0.3573
Bea	11.41 YES	3.71	3.71	0.3247
Noah	44.18 NO	28.77	0	0.6511

Each row illustrates one behavior from Section 8. Alice is partially filled: her cheap 20%–30% slices refunded, so her effective price rose from her 0.2933 receipt average to 0.3573 — within her own range bound $[P(20\%), P(40\%)] = [0.20, 0.40]$, and in fact within the 30%–40% sub-range that actually matched. Bea is prorated: she keeps half her shares at exactly her original 0.3247 average, because proration scales shares and cost together — quantity changed, price did not. Noah is the scarce side everywhere he traveled and fills completely.

The conservation identity holds exactly (four-decimal values; tables above are rounded to two):

$$E = 28.7682 + 7.4108 + 28.7682 = 64.9472$$

$$R = 17.0585 + 3.7054 + 0 = 20.7639$$

$$L = 22.8259 + 21.3574 = 44.1833 = F$$

$$R + L = 64.9472 = E$$

With a graduation threshold of 40 complete sets, $F = 44.18 \geq 40$ and the market graduates: 44.18 YES tokens and 44.18 NO tokens are minted against 44.18 units of locked collateral, and 20.76 units flow back as refunds. Settlement is then a closed pool:

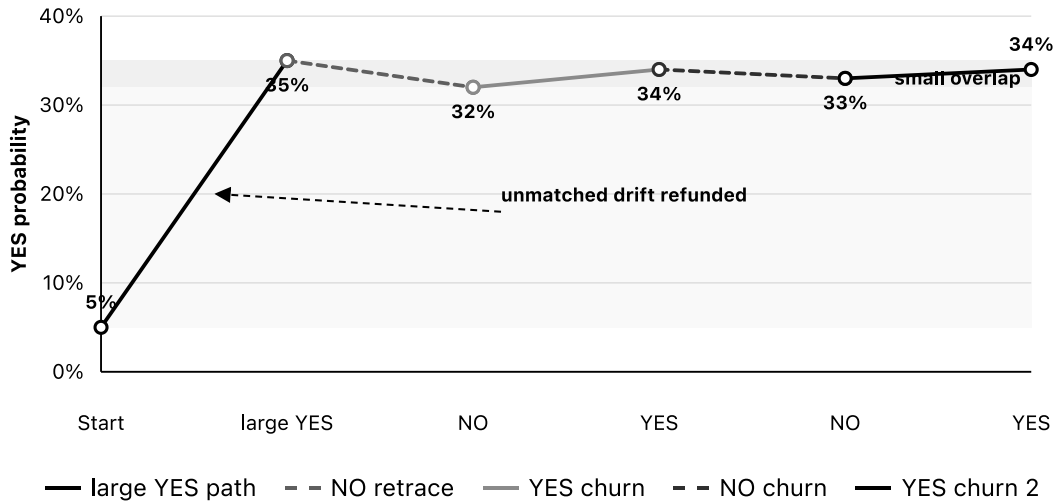
Resolution	Alice net	Bea net	Noah net	Sum
YES resolves true	+21.06	+7.71	−28.77	0
NO resolves true	−11.71	−3.71	+15.42	0

Net results sum to zero in both rows because the graduated market is exactly the locked pool — winners are paid entirely by losers' retained cost, and refunded segments sit outside the bet. Note that Alice's worst case is losing 11.71, her retained cost, not her 28.77 escrow: the unfilled majority of her intent was already safely refunded. Had the threshold been 50 instead, the same book would not graduate and all three traders would be refunded in full.

Example B: Large Drift Is Not Backfilled

Now suppose a market opens at 5% and a large YES intent moves it to 35%. Afterward, traders churn near the new price: NO moves 35% to 32%, YES moves 32% to 34%, NO moves 34% to 33%, and YES moves 33% to 34%.

Example B Price Path: Large Drift, Small Overlap



Churn near the new price clears only the bands it actually retraces. The one-sided 5% to 32% drift below it is refunded at path cost, however valuable it was as price discovery.

The big move may be genuinely informative — the market clearly no longer believes 5%. But information is not collateral. Below 32%, the large receipt's path is one-sided, so at minimum that whole segment refunds at its recorded cost:

$$\text{refund}_{\text{large YES}} \geq \int_{r_{0.05}}^{r_{0.32}} P_{\text{yes}}(r) dr = 100 \ln \left(\frac{0.95}{0.68} \right)$$

while the receipt's 32%–35% top competes pro-rata with the later YES churn for the small matched region. If that overlap meets the threshold, the market graduates carrying the discovered price; if not, everyone is refunded.

This is the mechanism's answer to paint-the-curve manipulation. A trader can always buy the displayed odds to any level, but drift without opposing flow converts at graduation into the trader's own refund — discovery is kept, unbacked exposure is not. Backfilling the drift would require the very thing the drift lacks: someone real on the other side.

10. Properties

No Subsidy

No funded loss budget exists anywhere in the lifecycle. Unmatched demand is refunded, never socialized.

Continuous Price Discovery

Every market quotes a smooth probability curve from its first block, at any opening prior, including deep longshots that reserve-based designs cannot quote without fake depth.

Fixed-Payout Discipline

The mechanism never drifts into pari-mutuel payoffs. Retained exposure is ordinary YES/NO complete-set exposure with one unit of collateral locked per set.

Price-Certain, Quantity-Uncertain Fills

Every slice of an intent fills or refunds at the price recorded when it traded. Other traders determine fill quantity; nothing determines fill price after the fact.

Cost-Basis Preservation

Clearing splits each receipt's escrow exactly: $\text{retained_cost} + \text{refund} = c$, per receipt, with no fees hidden in the identity.

Bounded Downside

The worst pre-graduation outcome is a full refund. Losses can only occur on filled shares after graduation, capped at retained cost.

Deterministic Clearing

Given the same frozen book, b , initial state, and rounding rules, every implementation produces identical bands, retained shares, locked collateral, and refunds.

Solvency By Construction

Locked collateral equals retained market cap ($L = F$) band by band, so maximum winner payout is fully funded before any token exists.

11. Limitations And Open Questions

Locked Capital

Receipts cannot be withdrawn, transferred, or sold before clearing. The lock is what makes displayed prices credible and clearing deterministic, but it is real illiquidity: capital is committed until graduation, cancellation, or the close-time refund.

No Guaranteed Fill

A receipt may fill fully, partially, or not at all, and the outcome depends on other traders' paths. The mechanism's promise is exactness, not certainty: every slice settles at its own recorded price, and the bounds of Section 8 always hold.

Refund Disappointment

A popular one-sided market can end in a large refund event. This is the mechanism working — unbacked exposure must not survive graduation — but users who mistake provisional receipts for fills will experience it as failure. Interfaces must label receipts as provisional and surface estimated retention and per-band exposure honestly.

Range Cognition

Buying an intent over a price range is a genuinely different object from buying tokens at a spot price, and no interface can fully hide that. The strip-of-limit-orders framing of Section 4 and the outcome table of Section 8 are the load-bearing explanations; they belong in the product, not just in this paper.

Graduation Gaming

An attacker may try to manufacture just enough path-compatible overlap to graduate a market on favorable terms. The clearing math is indifferent to who trades, so thresholds should be paired with non-price gates — unique-participant minimums, concentration limits, time-in-book requirements — before a minimally matched book is allowed to graduate.

Open Questions

1. How should b vary with market category, expected volatility, and opening prior?
2. What graduation threshold yields viable post-graduation liquidity without refunding too many genuinely useful small markets?
3. What rounding policy best preserves deterministic clearing under onchain integer arithmetic?
4. Which anti-manipulation gates earn their complexity in front of the threshold?
5. Can receipts be made transferable before clearing without reopening price-manipulation and refund-ownership problems?

None of these change the solvency argument. They determine how liquid, legible, and manipulation-resistant the bootstrap phase feels in practice.

12. Conclusion

A virtual LMSR is a strong bootstrapping surface for prediction markets. It quotes smoothly from the first trade, opens at any prior — including the 5% longshots that reserve designs cannot honestly quote — and costs the protocol nothing, because what it sells are priced intents rather than claims it cannot back.

Band-pass graduation clearing is what makes the free curve safe to build on. It matches only the demand that actually opposed itself — YES and NO flow crossing the same price bands in opposite directions — locks exactly one unit of collateral per retained complete set, and refunds every other path segment at the precise cost it recorded. The bettor's bargain is

stated, bounded, and honest: capital locked until clearing, fills partial and uncertain, prices exact, worst case a full refund.

The destination, throughout, is the standard market. Graduation hands fully collateralized CTF-style YES/NO tokens to ordinary prediction-market infrastructure and retires the scaffolding that produced them. The mechanism thus keeps three ideas separate that launch designs often conflate — probability discovery, fill, and settlement backing — and lets a market earn each one in order: born without a bankroll, priced from its first block, and graduated only when real opposing demand has paid for the exposure it keeps.

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